UI Design and Screentime: Human - Computer Interaction

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Background

This project seeks to explore intentional choices made by developers to enhance human-computer interaction (HCI). Through increased engagement, the time a user spends on their phone will increase. They become measures for the effectiveness of user interfaces' design of retaining their users.

User interfaces (UI) are "meticulously designed to attract and hold the user's attention" (Mujica), resulting in users' increased engagement with their screens. Some of these deliberate design choices include features such as "algorithmic content curation…visual interface design…the infinite scroll and the like button" (Mujica). In addition to these, 'dark patterns' are also used as "technologies hijack the human mind [24] and displease users". An example being Facebook offering guided settings to intentionally "keep users from certain settings" (Mildner and Savino). Keeping these features in mind could provide explanations for a user unknowingly spending more time on an application and social networking sites.

A user's willingness to dedicate their time is also important to consider when studying UI's effect. "Users are motivated to participate through features that prompt significant activities and contributions" to reward the user (Shawalludin et al.). Features contribute to "smartphone addictions" as they "maximise instant gratification" through "notifications" (Friedrichs et al.) or with a "satisfaction after winning a game" through achievements (Shawalludin et al.). Applications such as Tiktok also implement features like “streaks” that increase “frequent attempts to access social media” (Hillegass et al.). Studying the effects of design choices such as these allows developers to tackle the “conflict of interest between overall social good and revenue-driven businesses”, instead considering a “Human-Engaged Computing (HEC) framework” (Goethe et al.). This focuses on the well-being of the user along with a synergy between HCI.

Methodology

This research will measure the effects of certain UI decisions on users to determine how to help prevent higher levels of screentime. To collect information on different features’ effect on user usage, there will be a survey about the users’ screentime on a daily basis. In this survey, users will be questioned on specific features such as ‘infinite scroll’, ‘likes’, ‘rewards’, etc. and ratings for each feature's effectiveness for usage. With this information, users will be able to give which features keep them engaged with different platforms, evaluating the influence of UI on user engagement.

The first stage will consist of further features that are deliberately implemented as dark patterns or to enhance user experience. These features will be listed in the survey to rank in categories from strongly disagreeing to agreeing on what impacted their usage. After completing the survey, data will be overlooked to determine and rank the effects of certain features. Preferably these participants will be about 50 college students familiar with using technology for social reasons, school, work, etc. The final stage will be analyzing the data collected to provide a neat report on different features, ratings, and their relations to screentime. This information will be pieced together in a poster format to make the relationships easier to read to anyone interested in learning more about the topic.

Outcomes

With collected data from the surveys, tables will be made to visualize the effects of each feature with graphs to find deeper connections between them. Along with survey feedback, a poster will be made to include these visuals along with additional comments about each feature. Considering research, it is expected that features such as the infinite scroll and award systems will be the most mentioned by participants due to their frequent nature as social media platforms. User attentiveness to these features would also be important to consider to see whether these UI choices were as hidden as made out to be.

Significance

The significance of this research is to determine which features are most likely responsible for higher usage rates and screentime. Studying these results could help developers promote ethical engagement by adding features to help balance HCI. By knowing which features harm users the most, regulations could be set to protect usage. Obtaining input from a younger audience could be able to inform developers of the changing effects when it comes to UI and improve any harmful effects on users.

References

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Timeline

Dates	Tasks
January 1, 2027 - February 1, 2027	Research specific techniques and features used to enhance HCI. Create a survey based on this information.
February 1, 2027 - February 10, 2027	Find eligible and diverse participants for the survey who are willing to give their input through a detailed survey
February 10 , 2027 - February 15, 2027	Keep track of all participant information as they answer questions.
February 15 , 2027 - February 22, 2027	Overview of data collection, make sure everything is usable. - Make sure all participants answered survey questions and were able to provide accurate screentime. - “Accurate screentime” is active usage, not idling.
February 22 , 2027 - February 28, 2027	Take all data and organize it into a presentable format through a poster.

Budget

Participant incentives will include $10 amazon gift cards for anyone who fully completes the survey.

Item	Units and Cost	Total

Interviewee incentives	$10/person, 50 people	$500
Total	-	**$500**